# Cognition variables information.

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## Summary

### ***Cognitive function childhood, adolescence and early adulthood***

Cognitive function was first tested at age 8 years in a school setting, and again in this setting at ages 11 and 15 years. A test of verbal comprehension given at the latter age was then re-administered at age 26 years (with 10 additional items of increased difficulty to avoid a ceiling effect), as part of the first interview of the survey member as an adult.

COG8H, COG11H and COG15H are summary measures of test scores standardised to the whole population on each component test. If using a sub-sample of the data these scores must be re- standardised to the population being analysed.

### ***Cognitive function in adulthood***

Apart from the reading comprehension test which was repeated at age 26 years, cognitive function was not assessed again until age 43 years. At this time there was a shift away from tests of intellectual ability and towards the assessment of functional cognitive performance (memory, processing speed and motor praxis).

Some of these tests were repeated at age 53 years (1999), with the addition of tests measuring verbal ability, verbal fluency, prospective memory and delayed verbal memory. The NART is the National Adult Reading Test and allows a link back to related tests in childhood.

## Raw variables

### ***VOC8R, VOC11R***

Purpose : Score from a 50-item vocabulary test **at 8, 11 years** Year : 1954, 1957

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 50

99 Missing data

### ***VOC8N, VOC11N***

Purpose : Normalised score from a 50-item vocabulary test **at 8, 11 years** Year : 1954, 1957

**Definition**

Variables voc**8**r, voc**11**r normalised to a mean of 100 and SD 15

999 Missing data

### ***SC8R***

Purpose : Score from a 35-item verbal comprehension (sentence completion) test in 19**54 (at 8 years)** Year : 1954

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 30

99 Missing data

### ***SC8N***

Purpose : Normalised score from a 35-item verbal comprehension (sentence completion) test in 19**54 (at 8 years)** Year : 1954

**Definition**

Variable sc8r normalised to a mean of 100 and SD of 15

999 Missing data

## ***R8R, R11R***

Purpose : Score from a mechanical reading (pronunciation) test **at 8, 11 years** Year : 1954, 1957

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 50

99 Missing data

## ***R8N, R11N***

Purpose : Normalised score from a mechanical reading (pronunciation) test **at 8, 11 years** Year : 1954, 1957

**Definition**

Variable R8/11R normalised to a mean of 100 and SD of 15

999 Missing data

## ***PI8R***

Purpose : Score from a 60-item non-verbal ability (picture intelligence) test in 19**54 (at 8 years)** Year : 1954

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 60

99 Missing data

## ***PI8N***

Purpose : Normalised score from a 60-item non-verbal ability (picture intelligence) test in 19**54 (at 8 years)** Year : 1954

**Definition**

Variable PI8R normalised to a mean of 100 and SD of 15

999 Missing data

## ***COG8H***

Purpose : Standardised summary cognitive score in 19**54 (at 8 years)** Year : 1954

**Definition**

Summary variable of all standardised scores of variables VOC8R, SC8R, R8R, PI8R

**2. Specification**

COMPUTE COG8H=(VOC8H+SC8H+R8H+PI8H)/4 Where VOC8H, SC8H, R8H, PI8H are the test scores standardised to the whole population available on each test in the variables VOC8R, SC8R, R8R, PI8R. COMPUTE VOC8H=(VOC8R-16.28)/5.99 *Xh=(Xr-mean)/SD

## ***GA11R***

Purpose : Score from a 80-item verbal and non-verbal ability test in 19**57 (at 11 years)** Year : 1957

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 80

99 Missing data

## ***GA11N***

Purpose : Normalised score of the 80-item verbal and non-verbal ability test in 19**57 (at 11 years)** Year : 1957

**Definition**

Variable GA11R normalised to a mean of 100 and SD of 15

999 Missing data

## ***V11R***

Purpose : Score from a 40 verbal items test in 19**57 (at 11 years)** Year : 1957

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 40

99 Missing data

## ***V11N***

Purpose : Normalised score of the 40 verbal items in test v11r in 19**57 (at 11 years)** Year : 1957

**Definition**

Variable V11R normalised to a mean of 100 and SD of 15

999 Missing data

## ***NV11R***

Purpose : Score from a 40 non-verbal items test in 19**57 (at 11 years)** Year : 1957

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 40

99 Missing data

## ***NV11N***

Purpose : Normalised score of the 40 non-verbal items in test v11r in 19**57 (at 11 years)** Year : 1957

**Definition**

Variable NV11R normalised to a mean of 100 and SD of 15

999 Missing data

## ***A11R***

Purpose : Score from a 50-item arithmetic test in 19**57 (at 11 years)** Year : 1957

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 50

99 Missing data

## ***A11N***

Purpose : Normalised score of the 50-item arithmetic test a11r in 19**57 (at 11 years)** Year : 1957

**Definition**

Variable A11R normalised to a mean of 100 and SD of 15

999 Missing data

## ***COG11H***

Purpose : Standardised summary cognitive score in 19**57 (at 11 years)** Year : 1957

**Definition**

Summary variable of all standardised scores of variables VOC8R, SC8R, R8R, PI8R

**2. Specification**

COMPUTE COG11H=(R11H+VOC11H+GA11H+A11H)/4 Where R11H, VOC11H, GA11H, A11H are the test scores standardised to the whole population available on each test from the variables R11R, VOC11R, GA11R, A11R. COMPUTE R11H=(R11R-36.43)/10.51 *Xh=(Xr-mean)/SD

## ***GA15R***

Purpose : Score from the Heim AH4 Group Ability Test [a 130-item verbal and non-verbal ability test] in 19**61 (at 15 years)** Year : 1961

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 130

99 Missing data

## ***GA15N***

Purpose : Normalised score of the Heim AH4 Group Ability Test [a 130-item verbal and non-verbal ability test] in 19**61 (at 15 years)** Year : 1961

**Definition**

Variable GA15R normalised to a mean of 100 and SD of 15

999 Missing data

## ***V15R***

Purpose : Score from the 65 verbal items of the Heim AH4 Group Ability Test in 19**61 (at 15 years)** Year : 1961

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 65

99 Missing data

## ***V15N***

Purpose : Normalised score of the 65 verbal items of the Heim AH4 Group Ability Test in 19**61 (at 15 years)** Year : 1961

**Definition**

Variable V15R normalised to a mean of 100 and SD of 15

999 Missing data

## ***NV15R***

Purpose : Score from the 65 non-verbal items of the Heim AH4 Group Ability Test in 19**61 (at 15 years)** Year : 1961

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 65

99 Missing data

## ***NV15N***

Purpose : Normalised score of the 65 non-verbal items of the Heim AH4 Group Ability Test in 19**61 (at 15 years)** Year : 1961

**Definition**

Variable NV15R normalised to a mean of 100 and SD of 15

999 Missing data

## ***R15R***

Purpose : Score from the Watts Vernon Reading Test [a 35-item verbal comprehension (sentence completion) test] in 19**61 (at 15 years)** Year : 1961

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 35

99 Missing data

## ***R15N***

Purpose : Normalised score of the Watts Vernon Reading Test [a 35-item verbal comprehension (sentence completion) test] in 19**61 (at 15 years)** Year : 1961

**Definition**

Variable GA15R normalised to a mean of 100 and SD of 15

999 Missing data

## ***M15R***

Purpose : Score from a 47-item mathematics test in 19**61 (at 15 years)** Year : 1961

**Definition**

This variable is coded as a continuous variable with score values coded from 0 to 47

99 Missing data

## ***M15N***

Purpose : Normalised score of the 47-item mathematics test in 19**61 (at 15 years)** Year : 1961

**Definition**

Variable NV15R normalised to a mean of 100 and SD of 15

999 Missing data

## ***COG15H***

Purpose : Standardised summary cognitive score in 19**61 (at 15 years)** Year : 1961

**Definition**

Summary variable of all standardised scores of variables VOC8R, SC8R, R8R, PI8R

**2. Specification**

COMPUTE COG15H=(R15H+ GA15H+M15H)/3 Where R15H, GA15H, M15H are the test scores standardised to the whole population available on each test from the variables R15R, GA15R, M15R. COMPUTE R11H=(R15R-24.47)/6.64 *Xh=(Xr-mean)/SD

## ***WV26R***

Purpose : Score on Watts Vernon Reading Test (first 35 items) used at 15 years in 19**72 (at 26 years)** Year : 1972

**Definition**

Watts Vernon Reading Test used at 15 years (in 1961). This variable is coded as a continuous variable with score values coded from 0 to 35

99 Missing data

## ***R26R***

Purpose : Score on Watts Vernon Reading Test used at 15 in 19**72 (at 26 years)** Year : 1972

**Definition**

Watts Vernon Reading Test used at 15 years (in 1961), with an additional 10 items of increased difficulty to avoid a ceiling effect. This variable is coded as a continuous variable with score values coded from 0 to 45

99 Missing data

## ***R26N***

Purpose : Score on Watts Vernon Reading Test used at 15 in 19**72 (at 26 years)** Year : 1972

**Definition**

Variable R26R normalised to a mean of 100 and SD of 15

999 Missing data

## ***WV26H***

Purpose : Standardised score on Watts Vernon Reading Test (first 35 items) used at 15 years in 19**72 (at 26 years)** Year : 1972

COMPUTE WV26H=(WV26R-28.76)/5.4 *Xh=(Xr-mean)/SD

**R26H**

Purpose : Standardised score on Watts Vernon Reading Test used at 15 in 19**72 (at 26 years)** Year : 1972

COMPUTE R26H=(R26R-32.79)/8.26 *Xh=(Xr-mean)/SD

## ***ADULTHOOD (age 43 and 53) cognitive variables***

**mempr89, membp89, memlg89, memht89, memwt89, memac89, memcc89, memab89**

Purpose : Memory for 8 medical measures* taken in 1982 when interviewed in 1989 (at 43 years) Year : 1989

Definition Remembered that pulse rate* was measured at 1982 interview *pulse rate (pr), blood pressure (bp), lung function (lg), height (ht), weight (wt), arm circumference (ac), chest circumference (cc) and abdominal circumference (ac). This variable is coded as 0 Did not remember [...] measured 1 Remembered […] measured 7 No questionnaire 8 Not applicable 9 Unknown

**Specification**

DO IF (RANGE(BATCH89,1,18)) *if rcvd a 1989 questionnaire If (sysmis(menab89)) memab89=8 END IF

Note: these vars. have been subsequently labelled and an additional code ‘7’ No questionnaire added. MR’s original vars. were unlabelled and all those without a questionnaire in 1989 (N=2100) were blank (sysmis).

## Variables in this document

### 25 medium-confidence matches

| Variable | Label | How it was found |
|---|---|---|
| **Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Memory test [10135]** | | |
| memab89 | Remembered that abdominal circumference was measured at 1982 interview - at age 43 years | prose scan |
| memac89 | Remembered that arm circumference was measured at 1982 interview - at age 43 years | prose scan |
| membp89 | Remembered that blood pressure was measured at 1982 interview - at age 43 years | prose scan |
| memcc89 | Remembered that chest circumference was measured at 1982 interview - at age 43 years | prose scan |
| memht89 | Remembered that height was measured at 1982 interview - at age 43 years | prose scan |
| memlg89 | Remembered that lung function was measured at 1982 interview - at age 43 years | prose scan |
| mempr89 | Remembered that pulse rate was measured at 1982 interview - at age 43 years | prose scan |
| memwt89 | Remembered that weight was measured at 1982 interview - at age 43 years | prose scan |
| **Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Reading comprehension (Watts-Vernon) [10121]** | | |
| wv26r | Watts Vernon Reading Test (first 35 items) used at age 15 (edited raw score) - at age 26 years | prose scan |
| **Clinical data [51] › Cognitive function [101] › Childhood cognition [1011]** | | |
| cog11h | Cognitive function score (standardised to whole population on each test) - at age 11 years | prose scan |
| cog15h | Cognitive function score (standardised to whole population on each test) - at age 15 years | prose scan |
| cog8h | Cognitive function score (standardised to whole population on each test) - at age 8 years | prose scan |
| ga11n | Verbal and non-verbal ability - General Ability (80-items - normalised score) - at age 11 years | prose scan |
| ga11r | Verbal and non-verbal ability - General Ability (80-items - edited raw score) - at age 11 years | prose scan |
| ga15n | Heim A4 Group ability verbal and non-verbal test (130-items - normalised score) - at age 15 years | prose scan |
| ga15r | Heim A4 Group ability verbal and non-verbal test (130-items - edited raw score) - at age 15 years | prose scan |
| nv11n | Non-verbal intelligence test (40-items normalised score) - at age 11 years | prose scan |
| nv11r | Non-verbal intelligence test (40-items edited raw score) - at age 11 years | prose scan |
| nv15n | Heim A4 Group ability non-verbal test (65-items - normalised score) - at age 15 years | prose scan |
| nv15r | Heim A4 Group ability non-verbal test (65-items - edited raw score) - at age 15 years | prose scan |
| voc11n | Vocabulary test (50-items normalised score) - at age 11 years | prose scan |

| Variable | Label | How it was found |
|---|---|---|
| voc11r | Vocabulary test (50-items edited raw score) - at age 11 years | prose scan |
| voc8n | Vocabulary test (50-items normalised score) - at age 8 years | prose scan |
| voc8r | Vocabulary test (50-items edited raw score) - at age 8 years | prose scan |
| **Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]** | | |
| blood | Have you had anaemia in the last 10 years - at age 43 years | prose scan |

## 22 low-confidence matches

| Variable | Label | How it was found |
|---|---|---|
| **Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Reading comprehension (Watts-Vernon) [10121]** | | |
| r26n | Watts Vernon Reading Test used at age 15 (normalised score) - at age 26 years | prose scan |
| r26r | Watts Vernon Reading Test used at age 15 (edited raw score) - at age 26 years | prose scan |
| **Clinical data [51] › Cognitive function [101] › Childhood cognition [1011]** | | |
| a11n | Arithmetic test (50-items - normalised score) - at age 11 years | prose scan |
| a11r | Arithmetic test (50-items - edited raw score) - at age 11 years | prose scan |
| m15n | Mathematics test (47-items - normalised score) - at age 15 years | prose scan |
| m15r | Mathematics test (47-items - edited raw score) - at age 15 years | prose scan |
| pi8n | Non-verbal ability - picture intelligence (60-items - normalised score) - at age 8 years | prose scan |
| pi8r | Non-verbal ability - picture intelligence (60-items - edited raw score) - at age 8 years | prose scan |
| r11n | Mechanical reading - pronunciation - (normalised score) - at age 11 years | prose scan |
| r11r | Mechanical reading - pronunciation - (edited raw score) - at age 11 years | prose scan |
| r15n | Watts Vernon Reading Test (35-items verbal comprehension normalised score) - at age 15 years | prose scan |
| r15r | Watts Vernon Reading Test (35-items verbal comprehension edited raw score) - at age 15 years | prose scan |
| r8n | Mechanical reading - pronunciation - (normalised score) - at age 8 years | prose scan |
| r8r | Mechanical reading - pronunciation - (edited raw score) - at age 8 years | prose scan |
| sc8n | Verbal comprehension - sentence completion (35-items normalised score) - at age 8 years | prose scan |
| sc8r | Verbal comprehension - sentence completion (35-items edited raw score) - at age 8 years | prose scan |
| v11n | Verbal Intelligence (40-items - normalised score) - at age 11 years | prose scan |
| v11r | Verbal Intelligence (40-items - edited raw score) - at age 11 years | prose scan |
| v15n | Heim A4 Group ability verbal test (65-items - normalised score) - at age 15 years | prose scan |
| v15r | Heim A4 Group ability verbal test (65-items - edited raw score) - at age 15 years | prose scan |
| **Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Musculoskeletal system conditions [51016]** | | |
| back | In the last 12 months, have you had sciatica, lumbago or severe backache at age 53 years | prose scan |

| Variable | Label | How it was found |
| --- | --- | --- |
| **Questionnaire data [55] › Health and medical history [510] › Physical capability [5108]** | | |
| arm | Do you have difficulty because of long term health problems using either arm to reach up high above your head or to reach behind to tuck a shirt in or do up a zip?- at age 53 years | prose scan |